

PATENT LANDSCAPE REPORT ON NITRILE BUTADIENE RUBBER (NBR) SYNTHESIS, COMPOSITION, AND PROCESSING

1. ABSTRACT

This report synthesizes six primary patents relevant to nitrile butadiene rubber (NBR) and closely related NBR-based compositions, with emphasis on synthesis, preparation, compounding, and performance control. The landscape spans both direct NBR material design and downstream formulations that use NBR as a key elastomeric component. Across the set, the patents address different target embodiments: semiconductive rubber rollers and electro-conductive members that incorporate NBR as a blend or domain rubber, sealing rubber materials based on NBR/PVC blends, NBR compositions for rubber-metal laminates, highly saturated nitrile rubber compositions built from hydrogenated NBR fractions, and hydrogenated NBR compositions tuned for oil/fuel resistance and low-temperature performance.

The disclosed polymerization and preparation approaches are diverse. Two patents focus on conventional NBR composition engineering rather than monomer polymerization: one uses NBR blended with PVC and sulfur vulcanization, while another uses NBR with tailored carbon black packages and oven vulcanization for rubber-metal laminates. The highly saturated nitrile rubber patent discloses emulsion polymerization of acrylonitrile/butadiene-based latexes, followed by hydrogenation, coagulation, drying, and compounding with polyamide resin in a twin-screw extruder. The hydrogenated NBR patent similarly relies on hydrogenated NBR feedstocks with specified acrylonitrile content and iodine number, compounded with ester plasticizer and organic peroxide, then press/oven vulcanized. The electro-conductive member patent is structurally focused, using NBR and SBR as phase-separated rubbers in matrix/domain architectures rather than describing NBR synthesis itself.

Major formulation trends include precise control of acrylonitrile content, hydrogenation level, carbon black type and loading, plasticizer level, and crosslinking system. Water appears prominently in the emulsion-polymerization and hydrogenation routes, including aqueous soap solutions, coagulation, washing, and latex handling, but is absent from the solvent-based or dry compounding patents except where used in evaluation or workup. Overall, the evidence indicates that NBR performance is most strongly tuned through monomer composition, hydrogenation state, and compounding architecture rather than through a single universal synthesis route.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8277692B2
- Patent Title: Semiconductive rubber composition and semiconductive rubber roller employing ...
- Assignee: Sumitomo Rubber Industries, Ltd.
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a semiconductive rubber composition and roller made by blending ethylene oxide-containing copolymer rubber with chloroprene rubber, and optionally NBR, followed by kneading, extrusion molding, steam-pan vulcanization, bonding, and finishing. It is a compounding and shaping disclosure rather than an NBR polymerization synthesis route. - Thiourea-based vulcanization accelerator: 0.5 to 1.5 parts by mass per 100 parts by mass of rubber content — Abstract/Claims - Guanidine-based vulcanization accelerator: 0.5 to 1.5 parts by mass per 100 parts by mass of rubber content — Abstract/Claims - Peroxide-based crosslinking agent: 0.5 to 2.0 parts by mass per 100 parts by mass of rubber content — Abstract/Claims - Epichlorohydrin-ethylene oxide copolymer rubber: 50 parts by mass — Example 1 - Chloroprene rubber: 50 parts by mass — Example 1 - Thiourea-based vulcanization accelerator: 1.0 part by mass — Example 1 - Guanidine-based vulcanization accelerator: 1.0 part by mass — Example 1 - Peroxide-based crosslinking agent: 1.0 part by mass — Example 1 - Vulcanization temperature: 160° C. — Example 1 - Vulcanization time: 1 hour — Example 1 - Roll body outer diameter: 20 mm — Example 1 - Thiourea-based vulcanization accelerator: 0.5 / 1.5 / 1.6 parts by mass — Examples 2/3 and Comparative Examples 1/2 - Guanidine-based vulcanization accelerator: 0.5 / 1.5 / 1.6 parts by mass — Examples 4/5 and Comparative Examples 3/4 - Peroxide-based crosslinking agent: 0.5 / 2.0 / 2.1 parts by mass — Examples 6/7 and Comparative Examples 5/6 - Epichlorohydrin-ethylene oxide copolymer rubber: 30 parts by mass; chloroprene rubber: 70 parts by mass — Example 8 - Epichlorohydrin-ethylene oxide copolymer rubber: 70 parts by mass; chloroprene rubber: 30 parts by mass — Example 9 - NBR: 25 parts by mass — Example 11 - NBR acrylonitrile content: 31.5% — Example 11 - Medium & Water Role: The core process is dry compounding and shaping of rubber ingredients, not aqueous polymerization. Water appears only in the steam-pan vulcanization step as steam; no aqueous NBR synthesis medium is disclosed. - acrylonitrile content: 31.5%

Medium & Water Role - The core process is dry compounding and shaping of rubber ingredients, not aqueous polymerization. Water appears only in the steam-pan vulcanization step as steam; no aqueous NBR synthesis medium is disclosed. - Core reaction medium: Dry compounding / melt-kneading followed by extrusion molding and steam-pan vulcanization; no aqueous polymerization medium disclosed for NBR itself.

acrylonitrile content - 31.5%

Relevant Experimental Evidence - Example 1: Baseline semiconductive rubber roller was made from 50/50 epichlorohydrin-ethylene oxide rubber and chloroprene rubber with 1.0 phr each of thiourea accelerator, guanidine accelerator, and peroxide crosslinker. - Examples 2-3: Changing thiourea accelerator content to 0.5 and 1.5 parts by mass was tested against comparative levels of 0.4 and 1.6 parts by mass. - Examples 4-5: Changing guanidine accelerator content to 0.5 and 1.5 parts by mass was tested against comparative levels of 0.4 and 1.6 parts by mass. - Examples 6-7: Changing peroxide crosslinker content to 0.5 and 2.0 parts by mass was tested against comparative levels of 0.4 and 2.1 parts by mass. - Example 11: Addition of 25 parts by mass of NBR with 31.5% acrylonitrile content reduced roller hardness further.

Technical Relevance This patent is relevant because it explicitly incorporates NBR into a semiconductive rubber composition and identifies its effect on roller hardness. Although not an NBR synthesis patent, it provides formulation guidance for NBR-containing elastomer systems used in electrophotographic components.

Patent 2

- Patent Number: US8846804B2
- Patent Title: NBR composition and sealing rubber material
- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an NBR/PVC sealing rubber material prepared by compounding high-nitrile NBR with NBR/PVC blend polymer, sulfur, and vulcanization accelerators, then kneading, molding, and vulcanizing. The focus is on formulation control to suppress insolubilized substances and improve ozone resistance, kneader ejectability, and compression set. - NBR polymer nitrile content: 42.5% — Example 1 - NBR polymer Mooney viscosity: 77.5 [ML1+4 (100°C)] — Example 1 - NBR polymer: 55 parts by weight — Example 1 - NBR/PVC blend polymer PVC content: 35% — Example 1 - NBR/PVC blend polymer: 45 parts by weight — Example 1 - Carbon black: 70 parts by weight — Example 1 - Zinc oxide: 5 parts by weight — Example 1 - Age resister: 2 parts by weight — Example 1 - Plasticizer (ethyl hexyl adipate): 30 parts by weight — Example 1 - Sulfur: 0.5 part by weight — Example 1 -

CBS: 2 parts by weight — Example 1 - TMTD: 2.5 parts by weight — Example 1 - Primary vulcanization temperature: 160 to 180° C. — Example 1 - Primary vulcanization time: 10 to 20 minutes — Example 1 - Secondary vulcanization temperature: 160° C. — Example 1 - Secondary vulcanization time: 1 hour — Example 1 - Fuel C immersion temperature: 40° C. — Example 1 - Fuel C immersion time: 5 days — Example 1 - Precipitate threshold: 0.05 ml or less — Example 1 - Precipitate threshold: less than 0.1 ml — Example 1 - First compression set test temperature: 100° C. — Description - First compression set test time: 70 hours — Description - Second compression set test temperature: 100° C. — Description - Second compression set test time: 15 hours — Description - PVC content range: 15.70 to 17.50 wt% — Abstract/Claims - Insolubilized substance-generating causative substance limit: 0 to 0.15 parts by weight per 100 parts by weight of polymer — Claims - Example 2 NBR/PVC blend polymer: 50 parts by weight — Example 2 - Example 3 NBR/PVC blend polymer: 55 parts by weight — Example 3 - Comparative Example 1 high nitrile polymer: 100 parts by weight — Comparative Example 1 - Comparative Example 2 high nitrile polymer: 60 parts by weight; NBR/PVC blend polymer: 40 parts by weight — Comparative Example 2 - Comparative Example 3 high nitrile polymer: 45 parts by weight; NBR/PVC blend polymer: 55 parts by weight — Comparative Example 3 - Medium & Water Role: The core process is dry compounding and vulcanization of NBR/PVC blends. Water appears only in the fuel-oil precipitation test and in the description of compression-set testing, not as a polymerization medium. - acrylonitrile content: 42.5%

Medium & Water Role - The core process is dry compounding and vulcanization of NBR/PVC blends. Water appears only in the fuel-oil precipitation test and in the description of compression-set testing, not as a polymerization medium. - Core reaction medium: Dry kneading / melt compounding followed by press and oven vulcanization; no aqueous polymerization medium disclosed.

acrylonitrile content - 42.5%

Relevant Experimental Evidence - Example 1: A sealing rubber formulation used 55 parts high-nitrile NBR and 45 parts NBR/PVC blend polymer with sulfur/CBS/TMTD curing. - Example 2: Increasing the NBR/PVC blend polymer to 50 parts maintained the same evaluation protocol. - Example 3: Increasing the NBR/PVC blend polymer to 55 parts was tested with added stearic acid and wax. - Comparative Examples 1-3: Varying the NBR/PVC blend ratio showed the importance of PVC content control for precipitation and compression set. - Comparative Examples 4-8: Addition of stearic acid and/or wax at different levels was used to probe insolubilized-substance formation.

Technical Relevance This patent is relevant because it directly defines an NBR-based sealing composition and quantifies the NBR/PVC blend architecture that governs fuel resistance and compression set. It is a formulation-focused NBR patent rather than a polymerization route, but it provides practical composition limits for NBR-containing sealing materials.

Patent 3

- Patent Number: US9688834B2

- Patent Title: NBR composition and rubber-metal laminate
- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an NBR composition for rubber-metal laminates, prepared by kneading NBR with MT carbon black, a second carbon black, sulfur, accelerators, silica, zinc oxide, stearic acid, and age resistor, then dissolving the compound in a toluene/methyl ethyl ketone mixed solvent to form a coating solution and vulcanizing it on metal. The emphasis is on balancing reinforcement and surface smoothness rather than NBR polymer synthesis. - NBR nitrile content: 34% — Example 1 - NBR: 100 parts by weight — Example 1 - MT carbon black: 150 parts by weight — Example 1 - Carbon black having iodine absorption number 18 mg/g and DBP absorption number 42 cm³/100 g: 40 parts by weight — Example 1 - Sulfur: 2 parts by weight — Example 1 - Tetrabenzylthiuram disulfide: 5 parts by weight — Example 1 - N-cyclohexyl-2-benzothiazolyl sulfonamide: 9 parts by weight — Example 1 - Silica: 30 parts by weight — Example 1 - Zinc oxide: 5 parts by weight — Example 1 - Stearic acid: 2 parts by weight — Example 1 - Age resistor: 5 parts by weight — Example 1 - Toluene to methyl ethyl ketone ratio: 9:1 — Example 1 - Solid concentration: 25% by weight — Example 1 - Dried thickness: 100 μm — Example 1 - Vulcanization temperature: 210° C. — Example 1 - Vulcanization time: 3 minutes — Example 1 - Contact pressure: 200 MPa — Example 1 - Press temperature: 150° C. — Example 1 - Evaluation time: 10 minutes — Example 1 - MT carbon black: 90 / 150 / 250 / 50 parts by weight — Example 4 - SEAST TA carbon black: 70 / 0 / 100 / 10 / 70 parts by weight — Example 4 - NBR: 100 parts by weight — Example 4 - Sulfur: 2 parts by weight — Example 4 - Tetrabenzylthiuram disulfide: 5 parts by weight — Example 4 - N-cyclohexyl-2-benzothiazolyl sulfenamide: 9 parts by weight — Example 4 - Silica: 30 parts by weight — Example 4 - Zinc oxide: 5 parts by weight — Example 4 - Stearic acid: 2 parts by weight — Example 4 - Age resistor: 5 parts by weight — Example 4 - MT carbon black: 150 parts by weight — Example 6 - SEAST TA carbon black: 40 / 40 / 40 / 10 / 70 / 10 parts by weight — Example 6 - N-cyclohexyl-2-benzothiazolyl sulfenamide: 9 / 5 / 12 / 9 / 9 / 9 parts by weight — Example 6 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - acrylonitrile content: 34%

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Mixed organic solvent coating solution (toluene/methyl ethyl ketone) followed by oven vulcanization; no aqueous polymerization medium disclosed.

acrylonitrile content - 34%

Relevant Experimental Evidence - Example 1: A coating solution from NBR and a mixed carbon black package was applied to metal and oven-vulcanized to form a rubber-metal laminate. - Example 4: Multiple carbon black ratios were compared to show the effect on flow resistance,

abrasion resistance, surface roughness, and adhesion. - Example 6: Alternative accelerator levels were tested while keeping the NBR and filler system constant. - Example 2: Increasing N-cyclohexyl-2-benzothiazolyl sulfonamide to 5 parts by weight was tested against Example 1. - Example 3: Increasing N-cyclohexyl-2-benzothiazolyl sulfonamide to 12 parts by weight was tested against Example 1.

Technical Relevance This patent is relevant because it uses NBR as the base polymer and specifies a composition window that supports oven vulcanization with good surface smoothness. It does not synthesize NBR, but it provides a practical NBR compounding strategy for gasket and laminate applications.

Patent 4

- Patent Number: EP2692788B1
- Patent Title: Highly saturated nitrile rubber composition and crosslinked rubber
- Assignee: Zeon Corporation
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of a highly saturated nitrile rubber composition by blending a carboxyl group-containing highly saturated nitrile rubber (a1), a highly saturated nitrile rubber (a2), and polyamide resin, then kneading at elevated temperature in a twin-screw extruder and compounding with carbon black, plasticizer, antioxidant, stearic acid, and peroxide crosslinker. The underlying rubber feedstocks are themselves produced by aqueous emulsion polymerization, hydrogenation, coagulation, and drying. - Ion exchanged water: 180 parts — Synthesis Example 1 - 10 wt% sodium dodecylbenzene sulfonate aqueous solution: 25 parts — Synthesis Example 1 - Acrylonitrile: 37 parts — Synthesis Example 1 - Mono n-butyl maleate: 6 parts — Synthesis Example 1 - t-dodecyl mercaptan: 0.5 part — Synthesis Example 1 - 1,3-butadiene: 57 parts — Synthesis Example 1 - Polymerization temperature: 5°C — Synthesis Example 1 - Cumen hydroperoxide: 0.1 part — Synthesis Example 1 - Polymerization time: 16 hours — Synthesis Example 1 - Hydroquinone aqueous solution: 0.1 part — Synthesis Example 1 - Residual monomer removal temperature: 60°C — Synthesis Example 1 - Hydrogen pressure: 3 MPa — Synthesis Example 1 - Hydrogenation temperature: 50°C — Synthesis Example 1 - Hydrogenation time: 6 hours — Synthesis Example 1 - Methanol: 2 volumes — Synthesis Example 1 - Vacuum drying temperature: 60°C — Synthesis Example 1 - Vacuum drying time: 12 hours — Synthesis Example 1 - Acrylonitrile units: 35.6 wt% — Synthesis Example 1 - Butadiene units (including saturated parts): 59.0 wt% — Synthesis Example 1 - Mono n-butyl maleate units: 5.4 wt% — Synthesis Example 1 - Iodine value: 7 — Synthesis Example 1 - Carboxyl groups: 3.1×10^{-2} ephr — Synthesis Example 1 - Polymer Mooney viscosity: 55 [ML1+4, 100°C] — Synthesis Example 1 - Ion exchanged water: 200 parts — Synthesis Example 2 - Sodium carbonate: 0.2 part —

Synthesis Example 2 - Fatty acid potassium soap: 2.25 parts — Synthesis Example 2 - Acrylonitrile: 42 parts — Synthesis Example 2 - t-dodecyl mercaptan: 0.45 part — Synthesis Example 2 - 1,3-butadiene: 58 parts — Synthesis Example 2 - Polymerization temperature: 5°C — Synthesis Example 2 - Cumen hydroperoxide: 0.1 part — Synthesis Example 2 - Hydroquinone aqueous solution: 0.1 part — Synthesis Example 2 - Aluminum sulfate aqueous solution: 3 wt% with respect to nitrile rubber — Synthesis Example 2 - Hydrogen pressure: 3.0 MPa — Synthesis Example 2 - Acetone concentration: 12% — Synthesis Example 2 - Vacuum drying temperature: 60°C — Synthesis Example 2 - Vacuum drying time: 12 hours — Synthesis Example 2 - Acrylonitrile units: 40.5 wt% — Synthesis Example 2 - Butadiene units (including saturated parts): 59.5 wt% — Synthesis Example 2 - Iodine value: 7 — Synthesis Example 2 - Polymer Mooney viscosity: 100 [ML1+4, 100°C] — Synthesis Example 2 - Ion exchanged water: 180 parts — Synthesis Example 3 - 10 wt% sodium dodecylbenzene sulfonate aqueous solution: 25 parts — Synthesis Example 3 - Acrylonitrile: 37 parts — Synthesis Example 3 - Methacrylic acid: 4 parts — Synthesis Example 3 - t-dodecyl mercaptan: 0.5 part — Synthesis Example 3 - 1,3-butadiene: 59 parts — Synthesis Example 3 - Polymerization temperature: 5°C — Synthesis Example 3 - Cumen hydroperoxide: 0.1 part — Synthesis Example 3 - Hydroquinone aqueous solution: 0.1 part — Synthesis Example 3 - Residual monomer removal temperature: 60°C — Synthesis Example 3 - Acrylonitrile units: 36 wt% — Synthesis Example 3 - Butadiene units (including parts not saturated): 61 wt% — Synthesis Example 3 - Methacrylic acid units: 3 wt% — Synthesis Example 3 - Iodine value: 9 — Synthesis Example 3 - Carboxyl groups: 3.5×10^{-2} ephr — Synthesis Example 3 - Polymer Mooney viscosity: 68 [ML1+4, 100°C] — Synthesis Example 3 - Kneading temperature: 220°C — Example 1 - Carboxyl group-containing highly saturated nitrile rubber (a1): 21 parts — Example 1 - Highly saturated nitrile rubber (a2): 49 parts — Example 1 - Nylon 12: 30 parts — Example 1 - MT Carbon: 40 parts — Example 1 - Tri-2-ethylhexyl trimellitate: 10 parts — Example 1 - 4,4'-di-(α,α -dimethylbenzyl)diphenylamine: 1.5 parts — Example 1 - Stearic acid: 1 part — Example 1 - 1,3-bis(t-butylperoxyisopropyl)benzene 40% product: 7 parts — Example 1 - Kneading temperature: 320°C — Example 2 - Nylon 6: same amount as Nylon 12 in Example 1 — Example 3 - Carboxyl group-containing highly saturated nitrile rubber (a1): 10.5 parts — Example 4 - Highly saturated nitrile rubber (a2): 59.5 parts — Example 4 - Carboxyl group-containing highly saturated nitrile rubber (a1): 24 parts — Example 5 - Highly saturated nitrile rubber (a2): 56 parts — Example 5 - Nylon 12: 20 parts — Example 5 - Medium & Water Role: Water is part of the core polymerization medium for the rubber feedstocks, and it is also used later for latex coagulation, washing, and workup. The final composition is then melt-kneaded without water in the compounding step. - acrylonitrile content: 35.6 wt%

Medium & Water Role - Water is part of the core polymerization medium for the rubber feedstocks, and it is also used later for latex coagulation, washing, and workup. The final composition is then melt-kneaded without water in the compounding step. - Core reaction medium: Aqueous emulsion polymerization and aqueous hydrogenation/latex handling for the rubber feedstocks; later melt compounding in a twin-screw extruder.

acrylonitrile content - 35.6 wt%

Relevant Experimental Evidence - Synthesis Example 1: Carboxyl group-containing highly saturated nitrile rubber (a1) was made by aqueous emulsion polymerization of acrylonitrile/butadiene/mono n-butyl maleate, hydrogenated at 3 MPa and 50°C, then coagulated and dried. - Synthesis Example 2: Highly saturated nitrile rubber (a2) was made by aqueous emulsion polymerization of acrylonitrile/butadiene, hydrogenated at 3.0 MPa, coagulated with aluminum sulfate, washed, and dried. - Synthesis Example 3: Methacrylic-acid-unit-containing highly saturated nitrile rubber was prepared by aqueous emulsion polymerization, hydrogenation, coagulation, and drying. - Example 1: A highly saturated nitrile rubber composition was prepared by twin-screw kneading a1/a2/Nylon 12 at 220°C and then adding carbon black, plasticizer, antioxidant, stearic acid, and peroxide. - Example 2: Raising twin-screw kneading temperature from 220°C to 320°C was tested for rollability and physical properties.

Technical Relevance This patent is highly relevant because it directly discloses NBR-derived feedstocks with explicit acrylonitrile contents and a full aqueous emulsion polymerization/hydrogenation route. It provides one of the clearest process windows for making highly saturated nitrile rubber materials from NBR precursors.

Patent 5

- Patent Number: US7772305B2
- Patent Title: Hydrogenated NBR composition
- Assignee: Nok Corporation
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a hydrogenated NBR composition made by compounding hydrogenated NBR with ester-based plasticizer, organic peroxide, and optionally polyfunctional unsaturated compound and fillers, then kneading, press-vulcanizing, and oven-vulcanizing. The hydrogenated NBR feedstocks are selected by acrylonitrile content and iodine number, but the patent is primarily a formulation and curing disclosure rather than a synthesis route for NBR itself. - Hydrogenated NBR acrylonitrile content: 25-44% by weight — Abstract/Claims - Hydrogenated NBR iodine number: 32-65 — Abstract - Hydrogenated NBR iodine number: 40-65 — Claim 1 - Di(butoxyethoxyethyl) adipate plasticizer: 2-23 parts by weight — Claim 1 - Organic peroxide: 0.5-10 parts by weight — Claim 1 - Hydrogenated NBR acrylonitrile content: 30-40% by weight — Claim 2 - Ester-based plasticizer: 7-17 parts by weight — Claim 2 - Organic peroxide: 4-8 parts by weight — Claim 2 - Polyfunctional unsaturated compound: 0.5-10 parts by weight — Claim 3 - Carbon black/white carbon: 20-150 parts by weight — Claim 4 - Secondary vulcanization temperature: 120 to 200° C. — Description - Secondary vulcanization time: 1 to 24 hours — Description - Kneading temperature: 170° C. — Comparative Example 1 - Kneading time: 20 minutes — Comparative Example 1 - Secondary vulcanization temperature: 160° C. — Comparative Example 1 - Secondary vulcanization time: 3

hours — Comparative Example 1 - Hydrogenated NBR (Zetpole 2020) acrylonitrile content: 36 wt% — Comparative Example 1 - Hydrogenated NBR (Zetpole 2020) iodine number: 28 — Comparative Example 1 - FEF carbon black: 32 parts — Comparative Example 1 - MT carbon black: 35 parts — Comparative Example 1 - Ester-based plasticizer: 15 parts — Comparative Example 1 - Antioxidant Anti CD: 1.5 parts — Comparative Example 1 - Antioxidant Anti MBZ: 1.5 parts — Comparative Example 1 - Zinc white: 2 parts — Comparative Example 1 - Organic peroxide Percumyl D: 6 parts — Comparative Example 1 - Hydrogenated NBR (Zetpole 2030L) acrylonitrile content: 36 wt% — Example 1 - Hydrogenated NBR (Zetpole 2030L) iodine number: 57 — Example 1 - Blend of Zetpole 2020 and Zetpole 2030L: 80/20 wt% — Example 2 - Blend iodine number: 34 — Example 2 - Blend of Zetpole 2020 and Zetpole 2030L: 60/40 wt% — Example 3 - Blend iodine number: 40 — Example 3 - Blend of Zetpole 2020 and Zetpole 2030L: 30/70 wt% — Example 4 - Blend iodine number: 48 — Example 4 - Ester-based plasticizer: 5 parts — Example 5 - Ester-based plasticizer: 10 parts — Example 6 - Ester-based plasticizer: 20 parts — Example 7 - FEF carbon black: 15 parts — Example 8 - White carbon: 25 parts — Example 8 - Silane coupling agent A172: 2 parts — Example 8 - White carbon: 43 parts — Example 9 - Silane coupling agent A172: 2 parts — Example 9 - Compression set at high temp: 8-13% — Comparative Examples 2-7 - Compression set at low temp: 31-72% — Comparative Examples 2-7 - TR-10: -24 to -40° C. — Comparative Examples 2-7 - Medium & Water Role: Water is not part of the disclosed compounding/vulcanization medium, but it appears in the test descriptions for low-temperature and oil-resistance evaluations. No aqueous NBR synthesis step is disclosed in the extracted evidence. - acrylonitrile content: 25-44% by weight

Medium & Water Role - Water is not part of the disclosed compounding/vulcanization medium, but it appears in the test descriptions for low-temperature and oil-resistance evaluations. No aqueous NBR synthesis step is disclosed in the extracted evidence. - Core reaction medium: Dry compounding / kneading followed by press vulcanization and oven vulcanization; no aqueous NBR polymerization medium disclosed in the examples.

acrylonitrile content - 25-44% by weight

Relevant Experimental Evidence - Comparative Example 1: Baseline hydrogenated NBR composition used Zetpole 2020 with 36 wt% AN content and iodine number 28. - Example 1: Replacing the base hydrogenated NBR with Zetpole 2030L (36 wt% AN, iodine number 57) was tested. - Examples 2-4: Blends of Zetpole 2020 and Zetpole 2030L were used to vary iodine number while keeping AN content at 36 wt%. - Examples 5-7: Ester-based plasticizer level was varied from 5 to 20 parts by weight. - Examples 8-9: Carbon black was partially or fully replaced by white carbon with silane coupling agent.

Technical Relevance This patent is relevant because it defines the acrylonitrile-content window for hydrogenated NBR feedstocks and shows how plasticizer, peroxide, and filler levels affect sealing performance. It is not a direct NBR synthesis patent, but it is important for NBR-derived elastomer formulation and property tuning.

Patent 6

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses an electro-conductive member with a matrix-domain electro-conductive layer built from two different diene rubbers, where one phase is NBR and the other is SBR, combined with electro-conductive particles and vulcanizing agents. The patent focuses on phase-separated rubber architecture, viscoelastic matching, and resistance stability rather than NBR polymer synthesis. - Difference of absolute values of SP values between first and second rubbers: 0.2 to 4.0 (J/cm³)^{0.5} — Claims - $\tan \delta_1 / \tan \delta_2$: 0.45 to 2.00 — Claims - Volume resistivity of matrix: $1.0 \times 10^8 \Omega \cdot \text{cm}$ or higher — Claims - Volume resistivity of domains: 1.0×10^1 to $1.0 \times 10^4 \Omega \cdot \text{cm}$ — Claims - Electro-conductive particles in domains: 50 to 150 parts by mass per 100 parts by mass of second rubber — Claims - NBR (1) nitrile content: 15% — Example materials - NBR (2) nitrile content: 41.5% — Example materials - NBR (3) nitrile content: 27.5% — Example materials - NBR (4) nitrile content: 18.0% — Example materials - NBR (5) nitrile content: 35% — Example materials - NBR (6) nitrile content: 40.0% — Example materials - SBR (1) styrene content: 45% — Example materials - SBR (2) styrene content: 25% — Example materials - SBR (3) styrene content: 18% — Example materials - SBR (4) styrene content: 15% — Example materials - SBR (5) styrene content: 40% — Example materials - Hydrin SP value: 18.5 (J/cm³)^{0.5} — Example materials - Example 1 NBR: 100 parts by mass — Table 1 - Example 1 carbon black: 60 parts by mass — Table 1 - Example 1 zinc oxide: 5 parts by mass — Table 1 - Example 1 zinc stearate: 2 parts by mass — Table 1 - Example 1 sulfur: 3 parts by mass — Table 3 - Example 1 di-2-benzothiazolyl disulfide: 2 parts by mass — Table 3 - Example 1 tetramethylthiuram disulfide: 0.5 part by mass — Table 3 - Example 1 molding die inner diameter: 12.5 mm — Example 1 - Example 40 conditioning temperature: 23° C. — Example 40 - Example 40 conditioning relative humidity: 50% — Example 40 - Example 40 output speed: 75 sheets/minute — Example 40 - Example 40 output speed: 370 mm/sec — Example 40 - Example 40 image resolution: 1,200 dpi — Example 40 - Example 40 output count: 20,000 sheets — Example 40 - Example 40 printing rate: 1.0% — Example 40 - Example 40 stop time: 12 hours — Example 40 - Medium & Water Role: The disclosed process is dry compounding and molding of rubber compositions. Water appears only as a conditioning/environmental factor in testing (relative humidity) and is not part of the NBR preparation medium. - acrylonitrile content: 15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%

Medium & Water Role - The disclosed process is dry compounding and molding of rubber compositions. Water appears only as a conditioning/environmental factor in testing (relative humidity) and is not part of the NBR preparation medium. - Core reaction medium: Dry rubber compounding and molding; no aqueous polymerization medium disclosed.

acrylonitrile content - 15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%

Relevant Experimental Evidence - Example 1: An NBR-based domain composition and a separate rubber composition were prepared and used to form an electro-conductive member. - Examples 2-39: Multiple NBR/SBR, SBR/NBR, and other diene-rubber pairings were screened for matrix-domain structure and viscoelastic balance. - Example 40: A transfer member was evaluated under high-speed printing conditions after conditioning at 23°C and 50% RH. - Comparative Example 3: Using NBR as the matrix and EPDM as the domain showed poor interfacial bonding and resistance uniformity. - Comparative Example 6: NBR/NBR pairing failed to form a matrix-domain structure, underscoring the need for dissimilar rubbers.

Technical Relevance This patent is relevant because it explicitly uses NBR as one of the two diene rubbers in a phase-separated electro-conductive layer and provides composition/property relationships for NBR-containing conductive members. It does not disclose NBR synthesis, but it is important for NBR application design and formulation.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
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US8277692B2	Sumitomo Rubber Industries, Ltd.	2012	The core process is dry compounding and shaping of rubber ingredients, not aqueous polymerization. Water appears only in the steam-pan vulcanization step as steam; no aqueous NBR synthesis medium is disclosed.	31.5%	Example 1: Baseline semiconductive rubber roller was made from 50/50 epichlorohydrin-ethylene oxide rubber and chloropre
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US8846804B2	Nok Corporation	2014	The core process is dry compounding and vulcanization of NBR/PVC blends. Water appears only in the fuel-oil precipitation test and in the description of compression-set testing, not as a polymerization medium.	42.5%	Example 1: A sealing rubber formulation used 55 parts high-nitrile NBR and 45 parts NBR/PVC blend polymer with sulfur/CB
US9688834B2	Nok Corporation	2017	No water-related process step disclosed in the extracted evidence.	34%	Example 1: A coating solution from NBR and a mixed carbon black package was applied to metal and oven-vulcanized to form

EP2692788B1	Zeon Corporation	2015	Water is part of the core polymerization medium for the rubber feedstocks, and it is also used later for latex coagulation, washing, and workup. The final composition is then melt-kneaded without water in the compounding step.	35.6 wt%	Synthesis Example 1: Carboxyl group-containing highly saturated nitrile rubber (a1) was made by aqueous emulsion polymer
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US7772305B2	Nok Corporation	2010	Water is not part of the disclosed compounding/vulcanization medium, but it appears in the test descriptions for low-temperature and oil-resistance evaluations. No aqueous NBR synthesis step is disclosed in the extracted evidence.	re 25-44% by weight	Comparative Example 1: Baseline hydrogenated NBR composition used Zetpole 2020 with 36 wt% AN content and iodine number
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US11347156B2	Canon Kabushiki Kaisha	2022	The disclosed process is dry compounding and molding of rubber compositions. Water appears only as a conditioning/environmental factor in testing (relative humidity) and is not part of the NBR preparation medium.	15%, 41.5%, 27.5%, 18.0%, 35%, 40.0%	Example 1: An NBR-based domain composition and a separate rubber composition were prepared and used to form an electro-c
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SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US10731020B2
- Patent Title: Rubber composition for vulcanization molding, process for manufacturing the ...
- Assignee: Matsumoto Yushi-Seiyaku Co., Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a rubber composition for vulcanization molding with many possible base rubbers, but NBR is only one listed option and not the central invention.

Supporting Patent 2

- Patent Number: US3297792A
- Patent Title: Blends of (1) graft copolymer of vinyl chloride onto chlorinated polyolefin, (2 ...
- Assignee: Monsanto Co
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)

- Technical Note: The patent is about multicomponent blends where a rubberlike acrylonitrile/diene copolymer is only one blend ingredient, so it is not primarily about NBR itself.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- All six patents are formulation- or application-driven rather than direct monomer-feed synthesis patents, but EP2692788B1 uniquely discloses the aqueous emulsion polymerization and hydrogenation route for NBR-derived feedstocks.
- Acrylonitrile content is explicitly specified in US8277692B2 (31.5% for the added NBR), US8846804B2 (42.5% for the high-nitrile NBR in Example 1), US9688834B2 (34% for the NBR in Example 1), EP2692788B1 (35.6 wt% for a1; 40.5 wt% for a2; 36 wt% for the methacrylic-acid variant), US7772305B2 (25-44 wt% in the claims; examples at 36 wt%), and US11347156B2 (multiple NBR grades from 15% to 41.5%).
- The most synthesis-relevant process conditions are in EP2692788B1: emulsion polymerization at 5°C, hydrogenation at 3 MPa and 50°C, coagulation, washing, and drying. US7772305B2 instead emphasizes downstream peroxide curing of hydrogenated NBR at 170°C press vulcanization and 160°C oven vulcanization.
- US8846804B2 and US9688834B2 both use sulfur-cured NBR compositions, but US8846804B2 combines NBR with PVC to control precipitation and compression set, whereas US9688834B2 focuses on carbon black architecture for reinforcement and surface smoothness in rubber-metal laminates.
- US8277692B2 and US11347156B2 are electrophotographic/application patents where NBR is a component in multi-rubber systems; both rely on compounding and phase/curing design rather than NBR polymerization. US11347156B2 spans a wider NBR grade range and explicitly links NBR/SBR pairing to matrix-domain behavior.
- Water use is most extensive in EP2692788B1, where it is part of the polymerization medium and later used for coagulation and washing. In US8846804B2 and US7772305B2, water appears only in testing/workup contexts, while US9688834B2 is essentially water-free in the disclosed processing route.

4. CONCLUSION

The evidence shows that NBR performance is controlled primarily through acrylonitrile content, hydrogenation level, and compounding architecture rather than through a single universal synthesis route. For direct NBR synthesis, EP2692788B1 provides the clearest aqueous emulsion polymerization and hydrogenation pathway, with acrylonitrile contents around 35.6-40.5 wt% in the disclosed highly saturated nitrile rubbers and explicit hydrogenation at 3 MPa and 50°C. For practical NBR formulations, the most useful parameters are the acrylonitrile windows in US7772305B2 (25-44 wt%) and the specific NBR grades in US8846804B2, US9688834B2, and US8277692B2. Recommended evidence-based synthesis/formulation targets are: use aqueous

emulsion polymerization for precursor NBR when synthesis is required; maintain acrylonitrile content in the mid-range for balanced oil resistance and low-temperature behavior; and tune downstream properties with controlled filler, plasticizer, and crosslinking systems. For sealing and laminate applications, sulfur-cured NBR/PVC or NBR/carbon-black systems are strongly supported, while electrophotographic applications favor NBR as one phase in a multi-rubber architecture rather than as a standalone polymer.

5. REFERENCES

- US8277692B2 | Semiconductive rubber composition and semiconductive rubber roller employing ... | Sumitomo Rubber Industries, Ltd. | US | 2012 | <https://patents.google.com/patent/US8277692B2/en>
- US8846804B2 | NBR composition and sealing rubber material | Nok Corporation | US | 2014 | <https://patents.google.com/patent/US8846804B2/en>
- US9688834B2 | NBR composition and rubber-metal laminate | Nok Corporation | US | 2017 | <https://patents.google.com/patent/US9688834B2/en>
- EP2692788B1 | Highly saturated nitrile rubber composition and crosslinked rubber | Zeon Corporation | EP | 2015 | <https://patents.google.com/patent/EP2692788B1/en>
- US7772305B2 | Hydrogenated NBR composition | Nok Corporation | US | 2010 | <https://patents.google.com/patent/US7772305B2/en>
- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>